IdiomX: A Bilingual Benchmark for Idiom Understanding, Generation, and Cross-Lingual Retrieval

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Abstract

Idiomatic expressions remain difficult for natural language processing because their meanings often cannot be derived directly from the individual words. This challenge becomes significantly harder in multilingual settings, where models must align figurative meaning across languages, where models must align figurative meaning across languages. Existing idiom resources are often limited in scale, contextual diversity, or multilingual support.

We introduce IdiomX, a large-scale English–Arabic benchmark for idiom understanding and cross-lingual retrieval. IdiomX is built through a reproducible multi-stage pipeline that combines lexical resource collection, controlled large language model enrichment, and structured validation to produce high-quality and semantically consistent examples.

The dataset contains 174,956 contextualized examples covering 12,823 English idioms, with aligned English–Arabic meanings, idiomatic and literal usage labels, and enriched linguistic metadata. It also includes quality-control measures such as deduplication, semantic validation, and leakage-safe splitting.

We evaluate IdiomX on three tasks: idiom detection, context-to-idiom retrieval, and Arabic-context-to-English-idiom retrieval. Results consistently show that contextual transformer models significantly outperform lexical baselines for idiom detection. For context-to-idiom retrieval, a hybrid retrieval architecture achieves 0.867 Top-1 accuracy and 0.894 MRR. For cross-lingual retrieval, fine-tuned multilingual embeddings achieve 64.86% Top-1 and 81.46% Top-5 accuracy.

These results establish IdiomX as a reusable benchmark for figurative language understanding, cross-lingual retrieval, and multilingual semantic modeling. Moreover the construction pipeline is language-agnostic and can be extended to other languages, enabling scalable multilingual idiom benchmarking.

Keywords: idioms; figurative language; bilingual benchmark; cross-lingual retrieval; Arabic NLP; large language models

I. Introduction

Idiomatic expressions are a well-known challenge for natural language understanding because their meanings are often non-compositional: they cannot be inferred directly from the meanings of their individual words. Expressions such as *kick the bucket* or *spill the beans* require contextual and cultural interpretation rather than literal reading. As a result, idioms remain difficult for computational models in tasks such as semantic interpretation, translation, retrieval, and contextual reasoning.

Although large language models and contextual embeddings have substantially improved language understanding, idiomatic language continues to be a frequent source of error. One reason is that existing idiom datasets remain limited in important ways. Many are relatively small, focused on a single task, or primarily monolingual. Resources such as VNC-Tokens [1], SemEval-2013 Task 5b [2], PIE [3], and MAGPIE [4] have made valuable contributions, but they do not fully address the need for large-scale, context-rich, bilingual benchmarks that support both monolingual and cross-lingual evaluation.

To address this gap, we introduce IdiomX, a large-scale bilingual dataset for idiom understanding and cross-lingual semantic retrieval. IdiomX contains 174,956 contextualized examples covering 12,823 English idioms, with aligned English–Arabic meanings, idiomatic and literal usage labels, and additional linguistic metadata. Each idiom is represented through multiple contextual realizations, which improves both diversity and evaluation realism.

A key strength of IdiomX is its structured and reproducible construction pipeline. The dataset combines curated idiom resources with controlled large language model enrichment and validation, allowing us to expand coverage while maintaining semantic consistency and data quality. This design also makes the pipeline modular and extensible to additional languages.

IdiomX supports three benchmark tasks: idiom detection, context-to-idiom retrieval, and Arabic-to-English idiom retrieval. Together, these tasks provide a unified evaluation setting for studying figurative language understanding in both monolingual and multilingual conditions.

Contributions

This paper makes the following contributions:

- We introduce IdiomX, a bilingual English–Arabic dataset containing 174,956 contextualized examples across 12,823 English idioms.
- We propose a reproducible and scalable construction pipeline that combines curated resources, controlled LLM-based enrichment, and structured validation.
- We provide rich semantic and linguistic annotations, including idiomatic vs. literal usage and aligned English–Arabic meanings.
- We define a unified benchmark covering three tasks: idiom detection, context-to-idiom retrieval, and cross-lingual Arabic-to-English idiom retrieval. We define a unified benchmark covering three tasks: idiom detection, context-to-idiom retrieval, and cross-lingual Arabic-to-English idiom retrieval.
- We provide strong baselines and detailed analyses showing that contextual modeling, hybrid retrieval, and task-specific fine-tuning are all important for idiom understanding.

II. Related Work

Research on idiomatic language understanding has evolved along two main directions:

(1) the construction of annotated datasets and
(2) the development of computational models for idiom detection and interpretation.

A. *Idiom Datasets*

Early work on idiom understanding focused on relatively small-scale annotated datasets. The VNC-Tokens dataset [1] and the SemEval-2013 Task 5b dataset [2] established standard benchmarks for distinguishing between idiomatic and literal usage. However, these datasets are limited in scale and contextual diversity. Subsequent efforts expanded both coverage and dataset size.

The PIE Corpus [3] introduced a larger collection of idiomatic expressions with contextual annotations, while MAGPIE [4] further increased scale and diversity. Despite these improvements, such datasets remain predominantly monolingual and are typically designed for a single task, most commonly idiom detection.

In addition, many existing datasets focus on specific idiom types, such as verb–noun constructions (e.g., "take place"), fixed expressions (e.g., "by and large"), or semi-compositional phrases (e.g., "spill the beans"), which limits their generalization across broader categories of figurative language.

Recent work has explored multilingual and cross-lingual idiom resources [5], [6]. While these approaches aim to capture semantic equivalence across languages, they often rely on direct translation and lack large-scale contextual examples, balanced idiomatic versus literal supervision, or explicit modeling of ambiguity.

B. *Models for Idiom Understanding*

A variety of computational approaches have been proposed for idiom understanding. Early methods relied on surface-level lexical features and statistical models, including n-gram representations and traditional classifiers.

More recent work has focused on neural architectures, particularly transformer-based models such as BERT [13] and its variants, which capture contextual semantics through self-attention mechanisms. These models have demonstrated strong performance in idiom detection and figurative language understanding tasks.

In multilingual settings, cross-lingual embedding models and multilingual transformers have enabled semantic alignment across languages. Approaches based on Sentence-BERT [17] and multilingual representation learning [6] have shown effectiveness in semantic similarity and retrieval tasks. However, their performance is highly dependent on the availability of large-scale, well-annotated datasets.

C. *Limitations of Existing Work*

Despite these advances, several key limitations remain:

(1) limited dataset scale,
(2) insufficient contextual diversity,
(3) lack of bilingual or cross-lingual alignment, and
(4) narrow task coverage focused on a single problem formulation.

D. IdiomX in Context

To address these challenges, we introduce IdiomX, a large-scale multilingual idiom benchmark that integrates multiple desirable properties within a single dataset. IdiomX provides aligned English–Arabic semantic representations, includes idiomatic, literal, and borderline usage examples, and supports multiple tasks including detection, context-to-idiom retrieval, and cross-lingual retrieval.

Unlike prior datasets such as PIE [3] and MAGPIE [4], IdiomX combines large-scale contextual coverage with controlled generation, semantic validation, and low duplication (reuse factor ≈ 1.04). In addition, IdiomX explicitly models linguistic ambiguity through borderline cases, enabling more realistic and challenging evaluation scenarios.

These properties position IdiomX as a comprehensive benchmark for studying idiomatic language understanding in both monolingual and multilingual contexts.

Table 1: Comparison of IdiomX with widely used idiom datasets in terms of scale, multilingual support, and task coverage.

Dataset	Lang	Idioms	Inst.	Bi	Ref
VNC-Tokens	EN	53	2,984	N	ACL Anthology
SemEval-2013 Task 5b [10]	EN	85	~4,350	N	ACL Anthology
IDIX [9]	EN	52	4,022	N	ACL Anthology
MAGPIE [8]	EN	~1,700	56,000	N	ACL Anthology
PIE-English [11]	EN	~1,200	~20,100	N	arXiv
IdiomX [23]	EN+AR	12,823	174,956	Y	Public Repository [1]

Table 1: Comparison of IdiomX with existing idiom datasets in terms of scale, multilingual support, and task coverage.

As shown in Table 1, IdiomX significantly surpasses existing datasets in both scale and coverage, providing 174,956

[1] [1] Dataset and code available at:
https://huggingface.co/datasets/aymansharara/IdiomX
https://github.com/aymanshar/idiomx-dataset

contextualized examples across 12,823 idioms. It is the only dataset that simultaneously combines large-scale contextual diversity, bilingual semantic alignment (English–Arabic), and support for multiple benchmark tasks within a unified framework. In contrast to prior resources, which are typically limited in size, monolingual, or designed for a single task, IdiomX enables comprehensive evaluation of idiomatic language understanding across multiple dimensions.

This comprehensive design positions IdiomX as a next-generation benchmark for idiom understanding and cross-lingual semantic modeling.

III. DATASET CONSTRUCTION

The IdiomX dataset was constructed through a multi-stage pipeline combining curated idiom sources with large language model (LLM)-based enrichment and structured validation.

Figure 1 illustrates the overall dataset construction process.

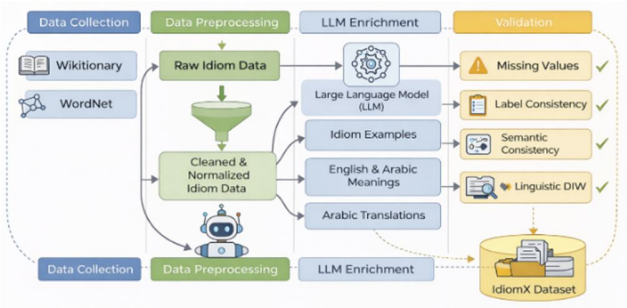

Figure 1 illustrates the overall dataset construction pipeline.

A. Data Sources and Collection

The construction process begins by aggregating idiomatic expressions from multiple linguistic resources, including publicly available idiom dictionaries and prior datasets such as VNC-Tokens [1], SemEval-2013 Task 5b [2], PIE [3], and MAGPIE [4]. This stage aims to maximize coverage across different types of idiomatic expressions, including verb–noun constructions, fixed expressions, and semi-compositional phrases.

Each collected expression is normalized into a canonical representation to ensure consistency. Duplicates and surface variations are removed, and each idiom is assigned a unique identifier (idiom_id) along with its canonical form (idiom_canonical), which is used throughout the dataset.

B. LLM-based Dataset Enrichment

To expand coverage and improve linguistic diversity, we perform large-scale dataset enrichment using the OpenAI GPT-4.1-mini model, a lightweight variant optimized for efficient high-volume generation [25].

The enrichment process is implemented as a batch-based asynchronous pipeline, where each idiom is converted into a structured prompt and processed at scale. This design ensures scalability, reproducibility, and efficient handling of large datasets.

For each idiom, the model generated:

- Natural example sentences
- Arabic translations of the examples
- English and Arabic meanings
- Bilingual explanations
- Usage labels (idiomatic, literal, borderline)
- Adversarial or challenging examples

C. Prompt Design

We design a structured prompt to enforce consistency, linguistic quality, and semantic correctness across all generated examples.

Key constraints included:

- The idiom must appear naturally within the sentence.
- Sentences must be grammatically correct and realistic.
- Arabic translations must be fluent and semantically accurate.
- Meanings must reflect the idiomatic interpretation.
- Usage labels must align with actual usage.

Additionally, adversarial examples are generated to introduce ambiguous or borderline cases, improving robustness for downstream evaluation.

This approach follows best practices in controlled LLM generation and instruction-based prompting [25].

D. Automated Validation and Correction

Following generation, all examples undergo a second-stage validation process using the same LLM in a structured verification mode.

Each row was evaluated according to strict criteria:

- Linguistic correctness (English and Arabic)
- Consistency between idiom and example
- Alignment between labels and meaning
- Accuracy of explanations
- Structural completeness of all fields

Validation outputs are enforced using a JSON schema, ensuring deterministic and machine-readable results [25].

Each entry was assigned one of the following statuses:

- **verified**: accepted without changes.
- **corrected**: modified to fix issues.
- **invalid**: rejected due to major issues

The pipeline supports checkpointing and incremental validation, enabling reliable processing of large-scale datasets.

E. Quality Control Strategy

Quality control was enforced through multiple layers:

1. **Pre-filtering**
   - Removal of non-idiomatic expressions.
   - Deduplication of examples.
2. **LLM Generation Constraints**
   - Strict prompt instructions.
   - Schema-constrained outputs.
3. **Post-generation Validation**
   - Automated correction of inconsistent entries.
   - Detection of semantic or grammatical errors.
4. **Statistical Validation**
   - Analysis of label distribution, length, and coverage.
   - Ensures dataset consistency and balance [26].

F. Examples of Dataset Quality

Idiomatic example
"He finally kicked the bucket after a long illness."
Meaning: To die.

Literal example
"She kicked the bucket across the yard."

Low-quality example (rejected or corrected)
"He kicked bucket yesterday very sadness."
Ungrammatical and semantically incorrect.

G. Final Dataset Characteristics

The final version of IdiomX contains 174,956 contextualized examples covering 12,823 English idioms, with 172,393 unique sentences and a reuse factor of approximately 1.04, indicating minimal duplication.

Each idiom is represented by an average of ~14 contextual examples, ensuring strong contextual diversity.

The dataset includes a balanced distribution of idiomatic and literal usages, together with a smaller set of borderline cases that capture linguistic ambiguity. In addition, all examples are enriched with aligned English–Arabic semantic annotations, including meanings, explanations, and usage labels.

These characteristics make IdiomX suitable for multiple tasks, including idiom detection, context-to-idiom prediction, and cross-lingual Arabic-to-English idiom retrieval

IV. DATASET ANALYSIS

In this section, we present a comprehensive analysis of the IdiomX dataset, examining its linguistic properties, quality, and distributional characteristics. The analysis highlights the dataset's suitability for robust idiom understanding and cross-lingual modeling.

A. **Compositionality Distribution**

Compositionality measures the extent to which the meaning of an idiom can be inferred from its constituent words. As shown in Fig. 2, the dataset is dominated by semi-opaque idioms (54.63%), followed by opaque expressions (33.38%) and transparent ones (11.99%). This distribution reflects realistic language usage, where most idioms exhibit partial compositionality rather than being fully opaque or fully transparent.

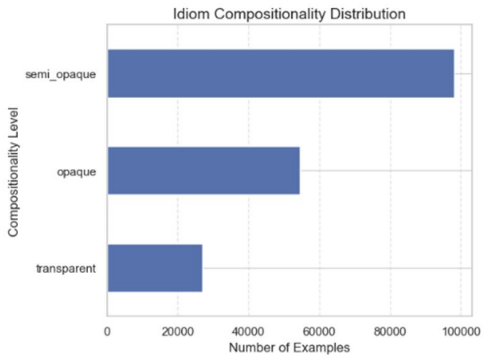

Figure 2 compositionality distribution.

B. **Register Distribution**

Register indicates the formality level of expressions. As shown in Fig. 3, neutral register dominates the dataset (59.9%), followed by informal usage (34.7%) and formal expressions (5.4%). This distribution ensures strong coverage of everyday conversational language while still including formal contexts, making the dataset applicable across diverse real-world scenarios.

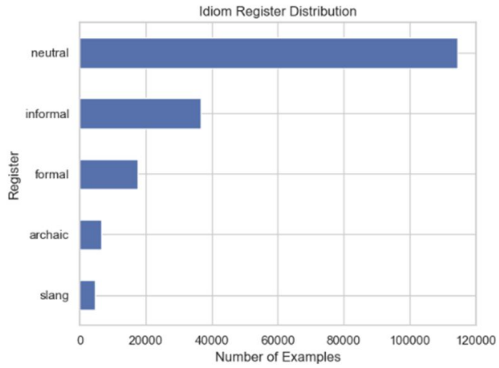

Figure 3 idiom register distribution.

C. **Difficulty Distribution**

Difficulty levels capture the learning complexity of idioms. As shown in Fig. 4, most idioms fall within intermediate and advanced levels, with fewer beginner-level expressions. This distribution reflects the inherently challenging nature of idiomatic language and ensures that the dataset supports realistic evaluation of model capabilities.

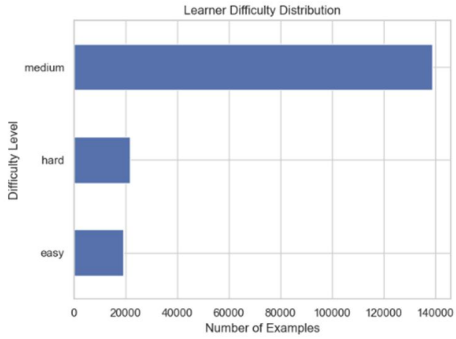

Figure 4 idiom learner difficulty distribution.

D. **Semantic Quality Distribution**

Semantic quality measures the correctness and coherence of generated examples. As shown in Fig. 5, over 77% of examples are rated as high quality, with a smaller proportion classified as medium or low quality. This confirms the effectiveness of the LLM-based generation and validation pipeline in producing reliable and meaningful data.

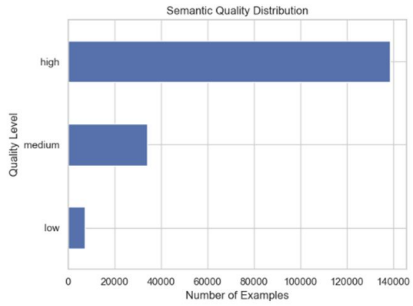

Figure 5 semantic quality distribution.

E. Sentence Length Distribution

Sentence length affects contextual complexity and model processing difficulty. As shown in Fig. 6, the dataset exhibits a well-balanced distribution of sentence lengths, covering both short and long contexts. This variability enhances the dataset's ability to support models across different levels of contextual reasoning.

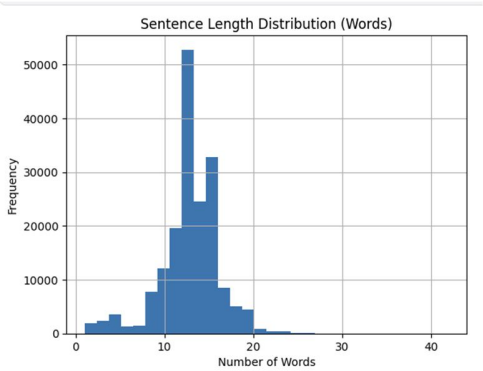

Figure 6 sentence length distribution.

F. Semantic Similarity Distribution

Semantic similarity captures how closely generated examples relate to the canonical idiom meaning. As shown in Fig. 7, the dataset includes a wide range of similarity levels, including high-similarity examples and more challenging borderline cases. This diversity is essential for training models capable of distinguishing subtle semantic differences.

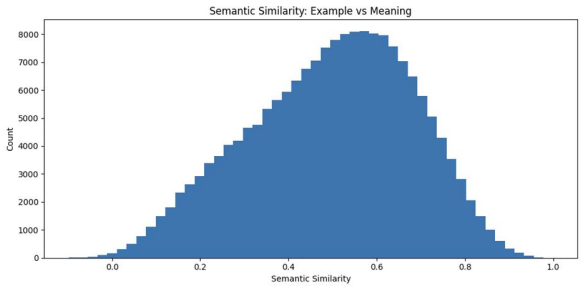

Figure 7 semantic similarity analysis.

G. *Idiomatic vs Literal Balance*

IdiomX maintains a balanced distribution between idiomatic and literal usage, ensuring that models are not biased toward a single interpretation. This balance is critical for classification and disambiguation tasks, where distinguishing between literal and figurative meaning is the primary objective.

H. *Contextual Diversity*

Each idiom is represented across multiple discourse types, including dialogue, narrative, formal, social media, question, and sarcastic contexts. This controlled variation ensures that models learn idioms in diverse linguistic environments rather than memorizing fixed patterns.

I. *Cross-Lingual Alignment*

All examples include aligned English and Arabic meanings and explanations, enabling cross-lingual evaluation. This bilingual structure supports research in multilingual representation learning and semantic transfer between languages.

J. *Key Dataset Strengths*

To summarize, IdiomX provides several key advantages over existing datasets:

• Large-scale coverage with 174,956 contextualized examples
• High contextual diversity with minimal duplication (reuse factor ≈ 1.04)
• Balanced idiomatic and literal usage
• High semantic quality with over 77% high-quality samples
• Structured generation ensuring consistency
• English–Arabic alignment for cross-lingual evaluation
• Inclusion of borderline cases for modeling ambiguity

These properties make IdiomX particularly suitable for modern NLP tasks requiring robustness, generalization, and multilingual understanding.

V. Benchmark Tasks

To comprehensively evaluate idiom understanding, we define three benchmark tasks that **capture** complementary aspects of figurative language processing, including classification, contextual reasoning, and cross-lingual semantic alignment.

Together, these tasks move beyond isolated evaluation settings and provide a unified framework for studying idioms across monolingual and multilingual scenarios.

A. Task 1: Idiom Detection

Idiom detection focuses on identifying whether an expression in a sentence is used idiomatically or literally, a core challenge in figurative language understanding [1], [2].

The task is formulated as a binary classification problem, where each input consists of a sentence containing a target expression. The model predicts whether the usage is idiomatic or literal.

This task evaluates a model's ability to distinguish figurative meaning from surface-level interpretation, which remains challenging even for modern contextual models..

B. Task 2: Context-to-Idiom Retrieval

Given a contextual sentence, the goal is to retrieve the most appropriate idiom that matches the underlying meaning.

We formulate this as a constrained retrieval problem over a predefined idiom inventory. Models must map contextual semantics to the correct idiomatic expression.

This task evaluates semantic understanding and expressive alignment, requiring models to capture intent beyond literal wording.

C. Task 3: Arabic-to-English Idiom Retrieval

This task extends retrieval to a cross-lingual setting, where the input is an Arabic sentence and the goal is to retrieve the correct English idiom.

It is formulated as a ranking problem, where candidate idioms are scored based on their semantic similarity to the input.

This task evaluates a model's ability to bridge meaning **across** languages, particularly in the presence of figurative expressions, which often lack direct lexical correspondence [6].

Together, these tasks form a progressive evaluation framework, moving from monolingual classification to

contextual reasoning and finally to cross-lingual semantic alignment.

VI. EXPERIMENTS AND RESULTS

This section presents the experimental evaluation of the IdiomX dataset across the defined tasks. The goal is to assess how well the dataset supports different types of problems, including idiom detection, context-to-idiom prediction, and cross-lingual retrieval.

We evaluate both classification and retrieval performance using standard metrics commonly used in NLP. The results provide an overall view of how models perform when dealing with idiomatic language, which often requires deeper contextual understanding.

A. Experimental Setup

The IdiomX dataset is split into training, validation, and test sets using a stratified approach. This ensures that both idiomatic and literal examples are evenly distributed across all splits.

Each idiom appears in both idiomatic and literal contexts, which helps create a balanced and fair evaluation setting.

For classification tasks, we report:

- Accuracy
- Precision, Recall, and F1-score

For retrieval tasks, we use:

- Top-1 accuracy
- Top-K accuracy

These metrics allow us to evaluate both exact predictions and ranking performance.

Transformer-based models are used across all tasks. In particular:

- BERT-based [13] models are used for classification tasks due to their strong contextual understanding.
- Multilingual models such as XLM-R are used for cross-lingual retrieval, as they are designed to handle multiple languages effectively [6]

All experiments are conducted using the same preprocessing pipeline and consistent training settings to ensure a fair comparison between models.

We begin our evaluation with the idiom detection task, which serves as the primary classification benchmark for assessing contextual understanding.

B. Task 1: Idiom Detection Using Transformer Models

1. Problem Definition

The goal of this task is to determine whether a sentence contains an idiomatic or a literal expression. Idioms are particularly challenging because their meanings cannot be directly inferred from the individual words.

We formulate this as a binary classification problem, where each sentence is labeled as either *idiomatic* or *literal*.

2. Models

Two transformer-based models were evaluated:

- **DistilBERT (Baseline)[12]:** a lightweight and efficient model used as a baseline.
- **RoBERTa (Improved Model)[14]:** a larger model designed to capture deeper contextual relationships in text.

Both models were fine-tuned on the same dataset using identical preprocessing and training settings to ensure a fair comparison.

3. Data Processing

Input sentences were tokenized using the pretrained tokenizer corresponding to each model. The following settings were applied:

- Padding to a fixed maximum length of 128 tokens
- Truncation of longer sequences

The dataset was split into training, validation, and test sets. The Hugging Face datasets library was used to handle preprocessing and data loading efficiently.

label	Count	Percentage (%)
literal	84,374	50.74
idiomatic	81,905	49.26

Table 2: Class Distribution of Idiomatic vs Literal Examples.

4. Training Setup

Both models were fine-tuned using the same configuration:

- Batch size: 8
- Learning rate: 2e-5
- Number of epochs: 3
- Evaluation performed at the end of each epoch
- Best model selected based on validation F1-score

This setup provided stable training while allowing a direct comparison between the two models.

5. Results

A. Validation Performance

RoBERTa showed consistent improvement across epochs, with a steady increase in F1-score and stable training behavior.

B. Test Performance

Model	Accuracy	Precision	Recall	F1-score
DistilBERT	0.8860	0.8804	0.9183	0.8884
RoBERTa	0.8959	0.8864	0.9055	0.8958

Table 3: Model Performance on Idiom Detection Task

RoBERTa outperforms the baseline across most metrics, particularly in precision and overall F1-score.

6. Training Behavior

The training curves show:

- A steady decrease in training loss
- A slight increase in validation loss in later epochs
- Continuous improvement in validation F1-score

This suggests stable convergence, with a small degree of overfitting toward the end of training.

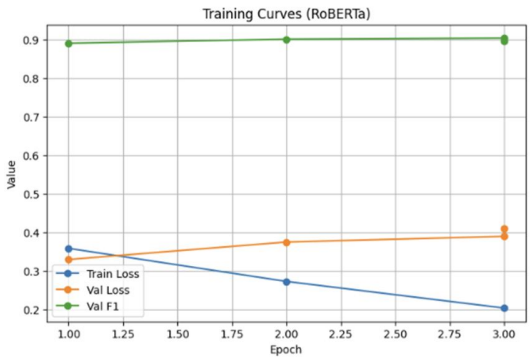

Figure 8: Training and validation loss curves across epochs.

As shown in Figure 8, training loss decreases steadily while validation loss slightly increases in later epochs.

7. *Error Analysis and Key Failure Modes*

To better understand model limitations, we analyze common failure cases.

(1) Limited or insufficient context

Short sentences or phrases without sufficient context make it difficult to determine whether the usage is idiomatic or literal.

(2) Strong idiom bias

Highly frequent idioms (e.g., "kick the bucket") are sometimes predicted as idiomatic even when used in a literal sense, indicating a bias toward memorized patterns.

(3) Context-dependent ambiguity

Certain expressions can be interpreted as either literal or idiomatic depending on subtle contextual cues, which are not always captured by the model.

(4) Indirect or implied meaning

Idiomatic meaning may be expressed implicitly without explicit lexical signals, making detection more challenging.

Overall, these errors suggest that the model relies heavily on surface patterns and may struggle when deeper contextual reasoning is required as also observed in prior work on figurative language understanding [3].

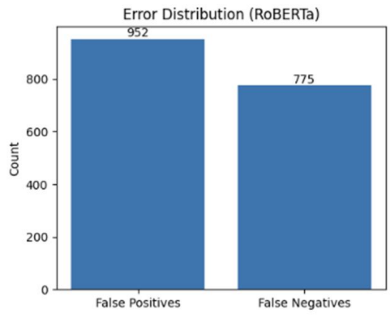

Figure 9: Error distribution (false positives vs false negatives).

As shown in Figure 9, false positives are slightly higher than false negatives, confirming the model's tendency to over-predict idiomatic usage.

8. *Qualitative Examples*

To further illustrate model behavior, we present representative examples covering both correct predictions and common failure cases:

Example 1 (Correct – Idiomatic)
Sentence:
"He finally kicked the bucket after a long illness."
Prediction: Idiomatic
Comment: The model correctly identifies the idiomatic meaning based on contextual cues.

Example 2 (Correct – Literal)
Sentence:
"She kicked the bucket across the yard."
Prediction: Literal
Comment: The physical action is explicit, and the model correctly classifies it.

Example 3 (Error – False Positive)
Sentence:
"He kicked the bucket lightly to move it aside."
Prediction: Idiomatic
Actual: Literal
Comment: The presence of a well-known idiom biases the model toward an idiomatic interpretation.

Example 4 (Error – False Negative)
Sentence:
"After years of stress, he finally let everything go."
Prediction: Literal
Actual: Idiomatic
Comment: The idiomatic meaning is subtle and not explicitly expressed, making it harder for the model to detect.

These examples show that most errors arise from ambiguity and context sensitivity rather than random misclassification.

9. *Confusion Matrix Analysis*

The confusion matrix shows generally balanced performance across both classes. However, the higher number of false positives confirms that the model leans slightly toward predicting idiomatic expressions.

The confusion matrix in Figure 10 illustrates the classification performance across both classes.

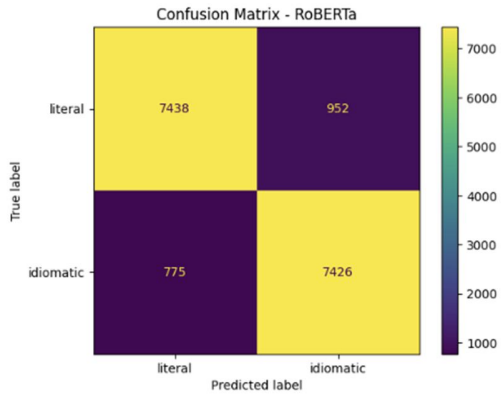

Figure 10: Confusion matrix for RoBERTa on the test set.

As shown in Figure 10, the model achieves strong performance across both classes, with high true positive rates for

both idiomatic and literal examples. A slightly higher number of false positives suggests a mild bias toward predicting idiomatic expressions, particularly for well-known phrases.

10. Ablation Study

The comparison between DistilBERT and RoBERTa highlights the impact of model capacity.

RoBERTa achieves higher precision and F1-score, indicating better contextual understanding. DistilBERT shows slightly higher recall, suggesting a tendency to generalize more aggressively.

This confirms that larger transformer models are more effective for capturing idiomatic meaning.

11. Precision–Recall Analysis

The precision–recall curve indicates that the model maintains high precision across most recall levels, with a drop only near maximum recall.

This reflects strong performance in identifying idiomatic expressions without introducing too many false positives.

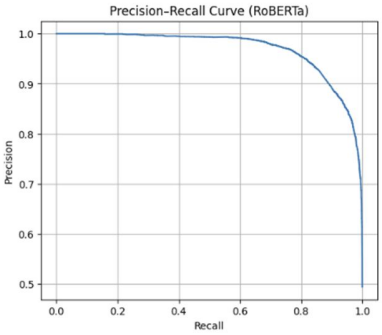

Figure 11: Precision–Recall curve for RoBERTa.

12. Conclusion

The results demonstrate that transformer-based models are well-suited for idiom detection, where understanding context is critical. While DistilBERT provides an efficient baseline, RoBERTa consistently achieves better performance, particularly in precision and F1-score.

Overall, the findings highlight that idiomatic language requires deeper contextual modeling, and that larger transformer architectures are more effective in capturing these nuances.

These results confirm that the IdiomX dataset provides a reliable foundation for idiom detection and serves as a strong benchmark for future research in figurative language understanding.

C. Task 2: Context-to-Idiom Retrieval

This task evaluates whether a model can recover the correct canonical idiom from an English contextual sentence that uses the idiom idiomatically. Unlike Task 1, which is a binary classification problem, Task 2 is formulated as a closed-set retrieval problem over a fixed idiom inventory.

Formally, given a contextual query x, the goal is to retrieve the correct idiom y* from a fixed idiom inventory I:

$$y^* = \text{argmax}_{y \in I} \text{score}(x, y)$$

where (s(x, i)) is the retrieval score assigned to idiom (i) given context (x).

This formulation requires both semantic understanding of the contextual sentence and precise matching to a canonical idiom form. Because the candidate inventory is fixed, the task provides a controlled and reproducible benchmark for evaluating retrieval-based idiom understanding.

1. Benchmark Construction and Split Strategy

The Task 2 benchmark is constructed from the IdiomX high-quality subset. To ensure a reliable retrieval setting, we keep only examples that satisfy the following conditions:

(1) the usage label is idiomatic,
(2) validation_status = valid, and
(3) semantic_quality = high.

After filtering, the final benchmark contains 38,604 examples associated with an idiom bank of 11,626 canonical idioms.

To prevent sentence leakage between training and evaluation, we apply a query-level split. Under this strategy, the same contextual sentence is never shared across splits. However, the same idiom may appear in both training and test sets through different contextual realizations, which is appropriate for the closed-set retrieval setting because the target inventory is fixed at evaluation time.

The final split contains 30,883 training examples and 7,721 test examples, with 5,356 idioms shared between train and test. This design ensures evaluation on unseen contexts while preserving realistic idiom coverage across both partitions.

Setting	Value
Dataset variant	IdiomX high-quality
Filtering	idiomatic only; valid; high semantic quality
Total examples	38,604
Idiom bank size	11,626
Train examples	30,883
Test examples	7,721
Shared idioms between train and test	5,356
Split strategy	Query-level
Sentence leakage across splits	None

Table 5: Task 2 benchmark statistics and split configuration.

2. Retrieval Architecture

We evaluate three retrieval settings of increasing complexity.

- **MiniLM dense retrieval**.

In the first setting, both contextual sentences and candidate idioms are encoded into a shared embedding space using MiniLM sentence embeddings [28]. Retrieval is then performed by ranking idioms according to semantic similarity between the input context and each idiom representation.

- **Hybrid retrieval.**

In the second setting, dense retrieval is combined with BM25 [27] sparse retrieval. The dense component captures semantic similarity beyond surface wording, while the sparse component captures direct lexical overlap between the context and idiom expressions. Their combination improves candidate recall by leveraging both semantic and lexical evidence.

- **Hybrid retrieval with reranking.**

In the final setting, a cross-encoder reranker [29] is applied to the top-10 candidates returned by the hybrid retriever. The reranker jointly scores each context–idiom pair, allowing more precise discrimination among semantically similar candidates. This produces the final ranked list used for evaluation.

This multi-stage design follows a standard retrieval pipeline in which early retrieval stages focus on candidate recall, while the final reranking [29] stage improves ranking precision.

3. Evaluation Metrics

Performance is evaluated using standard ranking metrics: Top-1 accuracy**,** Top-3 accuracy**,** Top-5 accuracy**,** and Mean Reciprocal Rank (MRR)**.**

Top-k accuracy measures whether the gold idiom y^*appears within the first k retrieved candidates. MRR evaluates the average reciprocal rank of the first correct prediction, providing a more sensitive measure of ranking quality.

The metrics subsection is fine as written, but if you want it to look slightly more formal, you can add one reference on ranking evaluation practice. This is optional.

4. Quantitative Results

Table 6 reports the main results for Task 2 under the closed-set retrieval setting. We compare three retrieval configurations: MiniLM dense retrieval, hybrid retrieval using MiniLM and BM25, and hybrid retrieval followed by cross-encoder reranking. MiniLM serves as the dense semantic baseline, while BM25 adds lexical matching signals that are known to remain useful in retrieval settings [17], [27]. The reranking stage further refines the top retrieved candidates using a cross-encoder scoring model, following standard multi-stage retrieval practice [29].

Model	Top-1	Top-3	Top-5	MRR
MiniLM	0.538	0.663	0.707	0.618
Hybrid (MiniLM + BM25)	0.763	0.871	0.901	0.825
Hybrid + reranker	0.867	0.922	0.926	0.894

Table 6: Main Task 2 retrieval results on the closed-set benchmark.

The results show a clear and consistent improvement across all evaluation metrics as the retrieval pipeline becomes more selective. MiniLM [28] achieves a Top-1 accuracy of 0.538 and an MRR of 0.618, indicating that dense retrieval alone provides a reasonable semantic baseline but still struggles with fine-grained ranking among semantically similar idioms. When BM25 is combined with MiniLM [28], performance increases substantially to 0.763 Top-1 and 0.825 MRR, showing that lexical matching and dense semantic similarity provide complementary evidence.

The strongest results are obtained with the full Hybrid + Reranker model, which reaches 0.867 Top-1, 0.922 Top-3, 0.926 Top-5, and 0.894 MRR. This confirms that the reranking stage is especially effective for resolving ambiguity among the strongest initial candidates. In other words, the hybrid retriever improves candidate recall, while the cross-encoder improves final ranking precision [29].

Figure 12 visualizes the overall comparison across Top-1, Top-3, Top-5, and MRR. The same trend is shown more clearly in Figure 13, where the Top-K retrieval curve demonstrates that each additional retrieval stage improves ranking quality, with the largest gains appearing at Top-1. This is important because exact canonical idiom recovery is the main objective of the task.

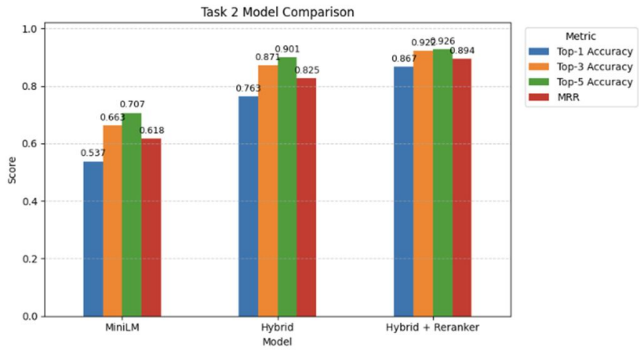

Figure 12: Performance comparison of Task 2 retrieval models across Top-1, Top-3, Top-5, and MRR.

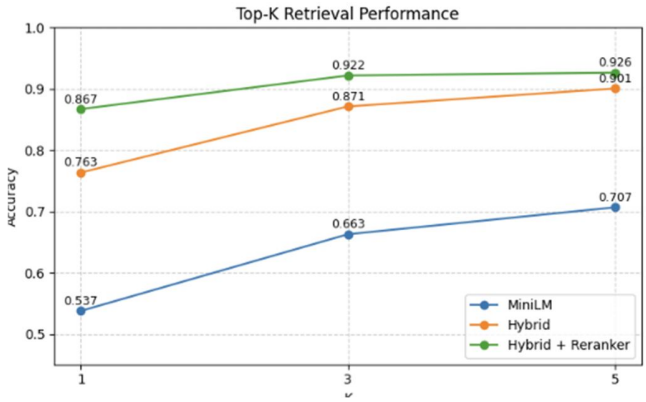

Figure 13: Top-K retrieval curves for Task 2 under the closed-set retrieval setting.

To quantify the improvement over the dense baseline, Table 7 and Figure 14 report the absolute gains relative to MiniLM [28]. The hybrid model improves Top-1 by **+0.226**, while the full reranking pipeline improves it by +0.329. Similar gains are observed for Top-3, Top-5, and MRR. These results indicate that context-to-idiom retrieval is not solved by semantic embedding alone; it benefits from combining semantic retrieval, lexical matching, and fine-grained pairwise reranking [17], [27], [29].

Metric	Hybrid Gain	Hybrid + Reranker
Top-1 Accuracy	+0.226	+0.329
Top-3 Accuracy	+0.209	+0.259
Top-5 Accuracy	+0.194	+0.220
MRR	+0.208	+0.276

Table 7: Absolute improvement over the MiniLM dense retrieval baseline.

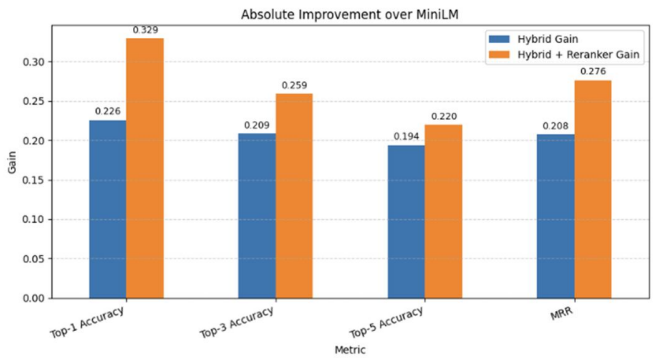

Figure 14: Absolute improvement over the MiniLM dense retrieval baseline on Task 2.

5. Error Analysis and Failure Modes

The best-performing model (Hybrid + Reranker) achieves strong performance, with 0.867 Top-1 accuracy and 0.894 MRR. However, a detailed analysis of the test set reveals 1,027 incorrect predictions out of 7,721 samples, as summarized in Figure 15. This corresponds to approximately 13.3% residual error, indicating that the task remains challenging despite high overall accuracy.

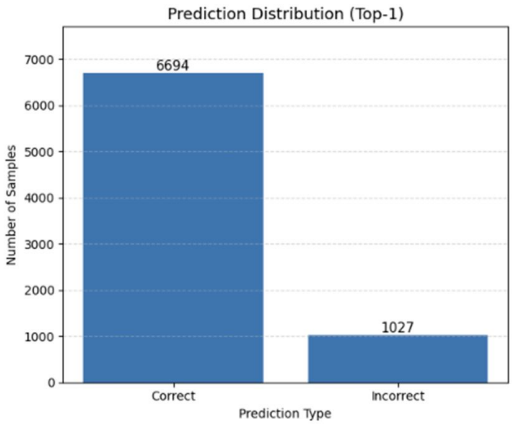

Figure 15: Distribution of correct and incorrect predictions for Task 2 on the test set.

A qualitative inspection of the errors reveals several consistent patterns.

First, the model often confuses idioms that are lexically or semantically similar, selecting plausible but incorrect expressions. Second, there is a noticeable frequency bias, where common idioms are preferred over less frequent but correct ones, a known effect in retrieval systems [27]. Third, the model sometimes relies on surface-level keyword matching, especially when lexical overlap is strong but semantically misleading. Finally, performance degrades on short or ambiguous contexts, where insufficient information is available to disambiguate between competing idioms.

Overall, these errors suggest that while the retrieval pipeline effectively combines semantic and lexical signals, idiom understanding still requires fine-grained contextual reasoning and improved handling of long-tail expressions [17], [29].

6. Qualitative Analysis

To better understand model behavior, we present qualitative examples from the test set illustrating both successful retrievals and typical failure cases. Table 8 shows representative examples produced by the best-performing model (Hybrid + Reranker).

The examples highlight that the model is generally able to recover the correct idiom when the contextual signal is strong and unambiguous. In such cases, the correct idiom is ranked at the top position, demonstrating effective semantic matching between the input context and the idiom meaning.

However, failure cases reveal consistent challenges. In particular, the model tends to select idioms that are lexically similar but semantically incorrect, or more frequent expressions that partially match the context. These errors confirm the patterns identified in Section 7, where lexical overlap and frequency bias can override deeper semantic understanding.

Overall, the qualitative results show that the retrieval pipeline performs robustly in clear contexts but struggles in cases requiring fine-grained semantic discrimination or involving less frequent idioms.

Table 8 presents representative examples from the best-performing model. The first two cases illustrate typical failure modes, while the final example shows a correct prediction under clear contextual conditions.

Query (shortened)	True Idiom	Top-1	Top-3 (shortened)	✓
"…eat them for breakfast."	eat someone for breakfast	don't worry	don't worry; eat someone…; take one…	✗
"Bone apple tea… bon appétit"	bon appétit	throw a bone to	throw a bone…; wise apple; contrary…	✗
"…finally broke the ice…"	break the ice	break the ice	break the ice; ease tension; start talking	✓

Table 8: Representative qualitative examples from Task 2

7. Task 2 Summary

In this task, we introduced a closed-set context-to-idiom retrieval benchmark built on the IdiomX high-quality dataset. The task requires mapping an idiomatic contextual sentence to its corresponding canonical idiom from a fixed inventory, combining semantic understanding with precise lexical grounding.

The results show that retrieval performance improves consistently across the proposed pipeline. Dense retrieval using MiniLM [28] provides a strong semantic baseline, while hybrid retrieval significantly improves performance by incorporating lexical matching signals [17], [27]. The addition of a cross-encoder reranker yields the best results, achieving 0.867 Top-1 accuracy and 0.894 MRR, demonstrating the importance of fine-grained pairwise scoring for resolving ambiguity among candidate idioms [29].

Overall, Task 2 highlights that idiom understanding can be effectively modeled as a retrieval problem, but requires a combination of semantic representations, lexical matching, and reranking to achieve high accuracy.

8. Discussion

The results of Task 2 provide several important insights into idiom understanding as a retrieval problem.

First, while dense semantic embeddings are effective at capturing contextual meaning, they are not sufficient on their own for precise idiom recovery. The strong improvement from hybrid retrieval confirms that lexical signals remain important, even in semantically driven tasks [17], [27]. This is particularly relevant for idioms, where surface forms often carry distinctive cues.

Second, the reranking stage plays a critical role in refining predictions. By jointly modeling the interaction between the context and candidate idioms, the cross-encoder improves ranking precision and resolves ambiguities among top candidates [29]. This demonstrates the effectiveness of multi-stage retrieval architectures for complex language understanding tasks.

Third, the error analysis reveals that remaining challenges are primarily due to semantic ambiguity, frequency bias, and limited contextual information. These findings suggest that

future improvements should focus on stronger contextual encoding, better handling of long-tail idioms, and training strategies that reduce reliance on frequency-based patterns.

Finally, the proposed benchmark provides a controlled and reproducible evaluation setting, enabling consistent comparison across retrieval architectures. As such, it can serve as a foundation for future research on idiom understanding, retrieval-based semantic modeling, and multilingual extensions.

D. Task 3: Arabic Context-to-Idiom Retrieval

This task evaluates whether a model can recover the correct canonical English idiom from an Arabic contextual sentence. Similar to Task 2, this is formulated as a closed-set retrieval problem over a fixed idiom inventory. However, unlike Task 2, this setting introduces a cross-lingual challenge, where the input is in Arabic and the target idiom is in English.

Formally, given an Arabic query x, the goal is to retrieve the correct idiom $y^*$from a fixed idiom inventory I:

$$y^* = \arg\max_{i \in I} sim(x, i)$$

where $\mathrm{sim}(x, i)$is the similarity score between the Arabic context and the candidate idiom.

This formulation requires both semantic understanding of the Arabic context and accurate alignment to a canonical English idiom. Compared to Task 2, the task is more challenging due to cross-lingual variation and differences in idiomatic expression across languages.

As in Task 2, we adopt a retrieval-based formulation because the candidate idiom inventory is fixed, enabling a controlled and reproducible evaluation setting.

1. Formulation

Each Arabic query $x$and candidate idiom $i$are mapped into vector representations using a shared encoder:

$$h_x = Encoder(x), \qquad h_i = Encoder(i)$$

Similarity is computed using cosine similarity [5]:

$$sim(x, i) = \frac{h_x \cdot h_i}{\|h_x\| \cdot \|h_i\|}$$

Prediction is defined as:

$$\hat{y} = \arg\max_{i} sim(x, i)$$

where cosine similarity [5] is used to rank candidate idioms in the shared embedding space.

2. Experimental Setup

To construct the Task 3 benchmark, we initially explored multiple approaches, including generation-based models (mT5 [19]), hybrid generation–retrieval variants, and Arabic semantic bank retrieval models. Early experiments used a broad Arabic input formulation that combined multiple fields and applied a strict split strategy to prevent overlapping at the idiom level.

While this setup was methodologically rigorous, it resulted in near-zero retrieval accuracy, indicating that the task formulation was overly constrained and not well aligned with a retrieval-based setting.

To address this, the task was reformulated as an example-level cross-lingual retrieval problem. The final benchmark retains only idiomatic examples and uses idiom_in_example_arabic as the Arabic query and idiom_canonical as the English retrieval target.

Instead of splitting by idiom identity, we apply an example-level split, allowing the same idiom to appear in both training and test sets through different contextual realizations. This setup evaluates whether the model can generalize to unseen Arabic contexts while preserving a realistic and fixed candidate inventory.

Under this revised formulation, retrieval performance improves substantially. The fine-tuned multilingual E5 model achieves strong results (Top-1 = 64.86%, Top-5 = 81.46%), demonstrating effective cross-lingual semantic alignment.

Overall, the results show that stronger pretrained multilingual embeddings alone provide only modest gains, while task-specific contrastive fine-tuning is the primary driver of performance improvement. Additional gains are obtained through richer supervision, such as multiple positive examples per query and extended training.

The final best-performing system is therefore the upgraded fine-tuned multilingual E5 retrieval model.

3. Candidate Representation

In Task 3, the candidate idiom inventory is represented using multiple Arabic textual views to improve semantic coverage. For each idiom, we include:

- example sentences (idiomatic usage)
- example meanings
- canonical meanings

Each representation is encoded into a shared multilingual embedding space, producing a dense representation matrix over the idiom inventory.

This multi-view representation allows the model to match Arabic queries against diverse contextual signals, improving robustness in cross-lingual retrieval.

4. Models Evaluated

We evaluate a set of multilingual dense retrieval models of increasing capability:

- Multilingual MiniLM [28] (baseline)
- Multilingual E5 (zero-shot)
- Multilingual E5 (fine-tuned)

MiniLM [28] serves as a lightweight semantic baseline. The E5 model introduces improved multilingual representation quality, while the fine-tuned variant is trained using contrastive learning on Arabic–idiom pairs to better align cross-lingual semantics.

While all models are evaluated, detailed analysis focuses on the best-performing configuration.

5. Quantitative Results

Performance is evaluated using standard ranking metrics, including Top-1 and Top-5 accuracy. Top-K accuracy measures whether the correct idiom appears within the top Kretrieved candidates.

$$Top - K = \frac{1}{N} \sum 1\{y \in TopK(x)\}$$

Table 8 reports the retrieval performance of the evaluated models.

Model	Embedding Dim	Top-1	Top-5
MiniLM-L12-v2 [28]	384	0.159	0.254
E5-base (zero-shot)	768	0.191	0.310
E5-base (finetuned)	768	0.578	0.761

Table 8: Retrieval performance on Task 3.

The results show a clear and consistent improvement across all models. MiniLM [28] provides a baseline for multilingual semantic retrieval but achieves limited performance, indicating that generic embeddings are insufficient for cross-lingual idiom understanding.

The E5 model improves performance under zero-shot settings, demonstrating the benefit of instruction-aware multilingual representations. However, the improvement remains modest, suggesting that pretrained embeddings alone are not sufficient for precise idiom retrieval.

The strongest results are obtained with the fine-tuned E5 model, which achieves 0.578 Top-1 and 0.760 Top-5 accuracy. This represents a substantial improvement over both baselines and highlights the importance of task-specific contrastive fine-tuning.

Overall, the results indicate that cross-lingual idiom retrieval requires not only strong multilingual representations but also explicit supervision to align Arabic contextual meaning with canonical idiomatic expressions.

Figure 16 illustrates the performance comparison across models, showing consistent gains at both Top-1 and Top-5.

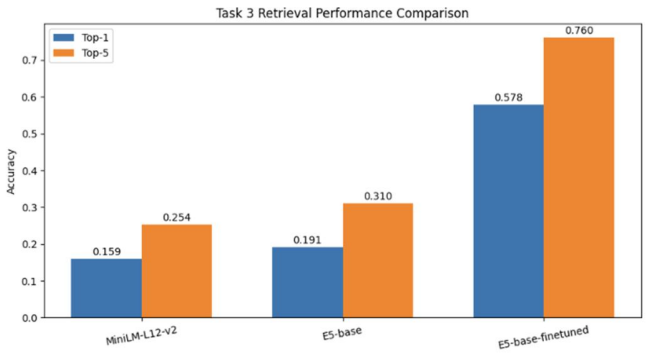

Figure 16: Model Retrieval Performance Comparison.

The results show a substantial improvement from fine-tuning, with E5-base-finetuned v2 outperforming all other models.

6. *Ablation Study and Model Progression*

The ablation study reveals a clear and consistent improvement across the retrieval pipeline, highlighting the impact of model architecture and training strategy.

As shown in Table 8, the MiniLM [28] baseline achieves limited performance, indicating that generic multilingual embeddings are insufficient for capturing idiomatic semantics in cross-lingual settings. Replacing MiniLM [28] with zero-shot multilingual E5 yields only modest gains, suggesting that pretrained multilingual representations alone are not sufficient for precise idiom alignment.

In contrast, fine-tuning the E5 model produces a substantial improvement, increasing Top-1 accuracy from 0.191 to 0.578 and Top-5 accuracy to 0.760. This represents a large relative gain over both baseline models.

These results confirm that task-specific contrastive training is the primary factor driving retrieval performance. Additional improvements are obtained through enhanced training strategies, including richer supervision and improved positive pair diversity, which contribute to better semantic alignment.

Overall, the ablation study demonstrates that idiom retrieval is not a general semantic similarity task, but a specialized problem that requires targeted supervision and fine-grained representation learning.

7. *Error Distribution Analysis*

To better understand model behavior, we categorize predictions into three outcome types:

- Top-1 correct: 6,662 queries (57.8%)
- Retrieved but misranked: 2,106 queries (18.3%)
- Fully incorrect: 2,762 queries (23.9%)

Figure 17 shows the distribution of these outcomes.

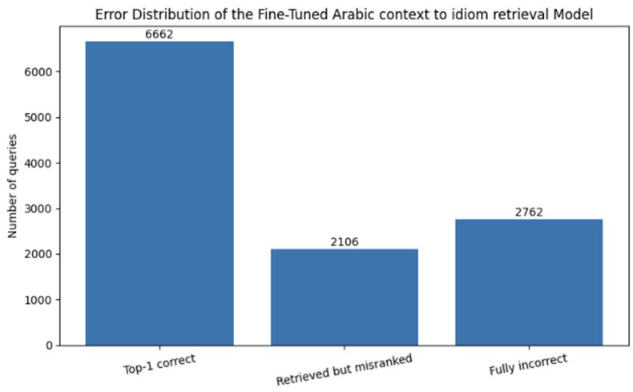

Figure 17: Error distribution across prediction outcomes.

A key observation is that a substantial portion of errors arises from ranking failures rather than retrieval failures. In many cases, the correct idiom is retrieved within the Top-5 candidates but is not ranked at the top position.

This indicates that the model successfully captures the overall semantic space but struggles with fine-grained discrimination between semantically similar idioms.

Fully incorrect predictions are typically associated with ambiguous Arabic inputs, weak contextual signals, or inherently difficult cross-lingual mappings.

Overall, the results suggest that improving ranking precision, rather than expanding retrieval coverage, is the primary direction for further performance gains.

8. *Similarity and Confidence Analysis*

We analyze the relationship between cosine similarity [5] scores and prediction correctness to assess model behavior and calibration.

Figure 18 shows that accuracy increases consistently with similarity score, indicating a strong positive correlation between similarity and prediction correctness. High-confidence predictions (similarity > 0.9) achieve very high accuracy, while low-similarity predictions are significantly less reliable.

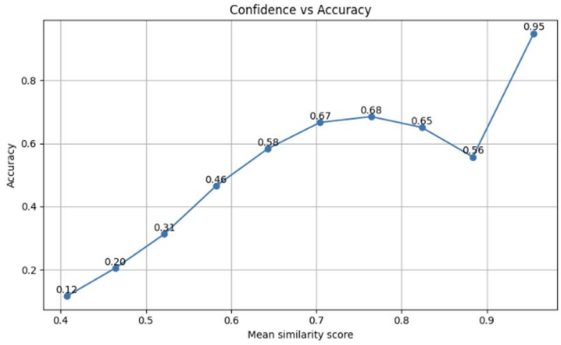

Figure 18: Accuracy as a function of similarity score.

Figure 19 further illustrates the distribution of similarity scores for correct and incorrect predictions. Correct predictions tend to have higher similarity values, while incorrect predictions are concentrated in lower score regions. However, there is a noticeable overlap between the two distributions.

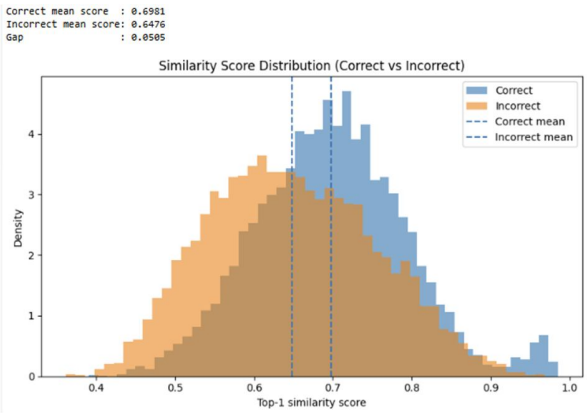

Figure 19: Similarity score distribution for correct and incorrect predictions.

This relatively small separation is also reflected in the mean similarity scores (0.698 for correct vs. 0.648 for incorrect), indicating that the embedding space is globally well-structured but locally dense.

The observed overlap suggests that many errors occur within semantically similar idiom clusters, rather than from complete semantic mismatches.

Overall, these findings show that cosine similarity provides a meaningful confidence signal but also highlight limitations in discriminative power when distinguishing between closely related idiomatic expressions.

9. *Threshold and Calibration Analysis*

We analyse model performance under varying similarity thresholds to evaluate its effectiveness as a confidence-based filtering mechanism.

Figure 20 shows that accuracy increases consistently as the similarity threshold becomes more restrictive. At lower thresholds, accuracy remains moderate due to the inclusion of ambiguous predictions, while higher thresholds (≥ 0.90) yield substantially higher accuracy, approaching near-perfect precision.

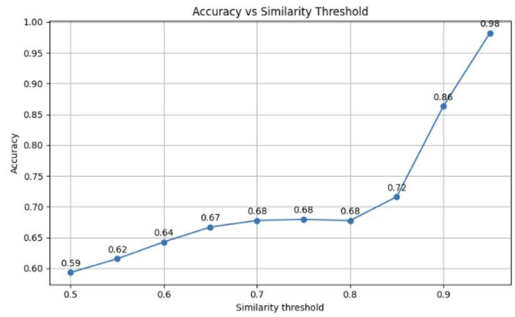

Figure 20: Accuracy as a function of similarity threshold.

This behavior reflects a clear trade-off between accuracy and coverage: stricter thresholds improve prediction reliability but reduce the number of accepted predictions.

These results indicate that similarity-based thresholding provides an effective calibration mechanism for practical deployment. By selecting an appropriate threshold, the system can balance precision and coverage, enable high-confidence predictions while deferring uncertain cases to fallback strategies.

10. *Recall@K Analysis*

We evaluate retrieval performance across different values of K to assess how often the correct idiom appears within the top-K candidates.

Figure 21 shows that recall increases rapidly for small values of K, with a large proportion of correct idioms retrieved within the Top-5 results. As K increases, the curve gradually saturates, indicating diminishing returns from retrieving additional candidates.

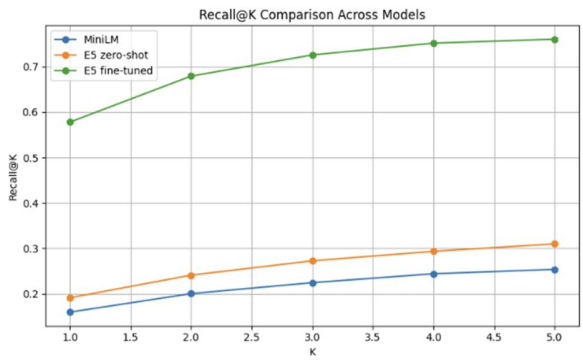

Figure 21: Recall@K performance across models.

This behavior suggests that the model is generally successful at identifying the correct semantic neighborhood, even when the correct idiom is not ranked at the top position.

In particular, the strong Recall@5 performance (0.760) indicates that the correct idiom is retrieved in most cases. However, the gap between Recall@1 and Recall@5 highlights persistent ranking limitations.

These findings reinforce earlier observations from the error analysis, confirming that the primary challenge lies not in retrieving relevant candidates, but in accurately ordering them within dense semantic clusters.

11. *Score Distribution Analysis*

Figure 22 compares the similarity score distributions across models. The fine-tuned E5 model produces a clear shift toward higher similarity values, indicating improved embedding alignment and stronger ranking confidence.

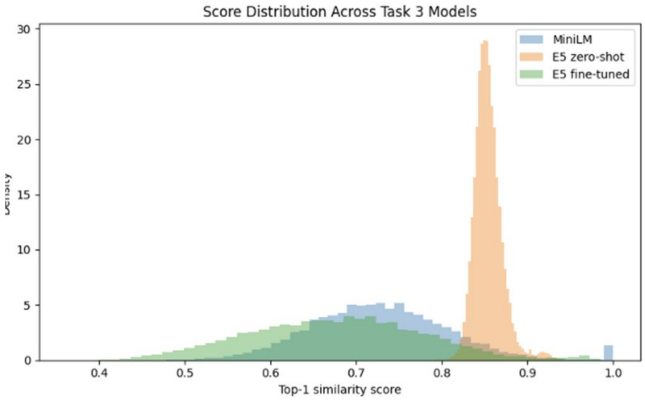

Figure 22: Similarity score distribution across models.

Compared to baseline models, the fine-tuned model exhibits better concentration of scores in higher similarity regions, reflecting improved semantic matching between Arabic contexts and idiomatic expressions.

This shift in distribution further confirms the effectiveness of task-specific fine-tuning in shaping a more discriminative embedding space.

12. *Qualitative Cross-Lingual Analysis*

To better understand model behavior, we present representative examples illustrating both successful predictions and typical failure cases.

The model performs well when the Arabic context provides clear semantic signals. For example:

- Arabic: "انتهى كل شيء في النهاية"
  Prediction: *come to an end*
- Arabic: "شارك همومك مع الآخرين لتخفيفها"
  Prediction: *a problem shared is a problem halved*
- Arabic: "لا توجد أي فرصة لحدوث ذلك"
  Prediction: *not a chance*

In these cases, the model successfully captures the underlying meaning and maps it to the appropriate idiomatic expression.

However, failure cases reveal consistent challenges:

- Arabic: "يجب أن تبطئ كلامك حتى يفهمك الجميع"
  Prediction: *please speak more slowly*
  Target: *slow down*

Here, the model retrieves a semantically related phrase rather than the canonical idiom, indicating sensitivity to surface-level similarity.

Across examples, successful predictions are typically associated with strong contextual signals and clear semantic intent, while errors often arise in ambiguous contexts or when multiple idioms share similar meanings.

Notably, incorrect predictions are often assigned high similarity scores, indicating overconfidence in dense semantic regions. This behavior aligns with earlier findings and highlights limitations in fine-grained ranking.

13. *Hard* Negative Analysis

Analysis of hard negative cases reveals that incorrect predictions are frequently semantically close alternatives to the correct idiom.

For example, idioms such as *in one fell swoop* vs. *at one stroke*, or *packed to the rafters* vs. *stuffed to the gills*, often appear within the same candidate set and are difficult to distinguish.

This indicates that the model struggles to differentiate between idioms within the same semantic cluster, where distinctions are subtle and context-dependent.

These findings suggest that incorporating hard negative mining during training could improve ranking precision by encouraging the model to better separate closely related idiomatic expressions.

14. *Discussion*

The results demonstrate that Arabic context-to-idiom retrieval is significantly more challenging than its monolingual counterpart due to cross-lingual semantic gaps and variation in idiomatic expression.

Unlike generation-based approaches, retrieval models rely on the structure of the embedding space to align meanings across languages. While multilingual models improve this alignment, they remain sensitive to subtle semantic differences and cultural variations in idiomatic usage.

The experimental results show that replacing the MiniLM [28] baseline with multilingual E5 leads to consistent improvements in retrieval performance. However, the gains from pretrained multilingual representations alone remain limited. The most substantial improvement is achieved through task-specific contrastive fine-tuning, which enables more precise alignment between Arabic contextual inputs and canonical English idioms.

Despite strong overall performance, the model continues to struggle with fine-grained semantic distinctions, particularly among idioms within the same semantic cluster. This highlights an inherent limitation of purely retrieval-based approaches for idiom understanding.

These findings suggest that future work may benefit from hybrid or generative approaches that can better capture subtle semantic nuances beyond nearest-neighbor retrieval.

15. *Task Summary*

In this task, we introduced a cross-lingual context-to-idiom retrieval benchmark, requiring the mapping of Arabic contextual inputs to canonical English idioms within a fixed candidate inventory.

The fine-tuned multilingual E5 model achieves the best performance, demonstrating strong capability in capturing cross-lingual semantic alignment. However, detailed analysis shows that most remaining errors arise from ranking ambiguity rather than retrieval failure.

This indicates that the primary challenge lies in distinguishing between semantically similar idioms within dense embedding regions.

Overall, the results highlight that effective cross-lingual idiom retrieval requires not only strong multilingual representations but also targeted training strategies that improve ranking precision.

These findings position cross-lingual idiom retrieval as a challenging and promising direction for multilingual semantic modeling, with clear opportunities for improvement through ranking optimization, hard negative mining, and enhanced contextual modeling.

These results highlight that cross-lingual idiom retrieval is not only a retrieval problem but also a semantic alignment challenge, where models must bridge figurative meaning across languages rather than rely on surface similarity.

VII. Discussion

The experimental results demonstrate that IdiomX supports a diverse range of NLP tasks related to idiom understanding, each requiring different modeling capabilities.

The findings highlight three key insights. First, idiom detection relies heavily on contextual modeling rather than surface-level lexical cues, confirming that idiomatic meaning cannot be inferred from isolated tokens. Second, context-to-idiom mapping is not purely a retrieval or generation problem, but requires precise semantic alignment between contextual meaning and canonical expressions. Third, hybrid approaches consistently outperform individual paradigms by combining generative flexibility with retrieval-based grounding.

Across tasks, a consistent pattern emerges: while retrieval-based models are effective at identifying the correct semantic neighborhood, they often struggle with fine-grained ranking among semantically similar idioms. In contrast, generative models offer greater flexibility but may lack precision in producing canonical idiomatic forms. This suggests that idiom understanding benefits from a division of labor, where generative models provide contextual reasoning and retrieval mechanisms ensure discrete and accurate grounding.

In the cross-lingual setting (Task 3), these challenges become more pronounced due to semantic variation across languages. Although multilingual embeddings enable alignment between Arabic contexts and English idioms, performance remains limited by ambiguity within dense semantic clusters. This highlights the importance of improved ranking strategies and training methods that better capture subtle semantic distinctions.

Overall, the results position idiom understanding as a complex semantic task that requires integrating contextual reasoning, structured retrieval, and cross-lingual alignment within a unified framework.

VIII. Applications

The IdiomX dataset supports a wide range of natural language processing tasks related to figurative language understanding, including:

- idiom detection
- context-to-idiom generation
- cross-lingual idiom mapping
- multilingual figurative language evaluation

The dataset is particularly well-suited for evaluating transformer-based models and large language models on non-compositional language, where meaning cannot be derived directly from surface forms.

IX. Dataset Availability

The IdiomX dataset is publicly available to support reproducibility and further research in idiomatic language understanding.

The dataset can be accessed through the following platforms:

Hugging Face:
https://huggingface.co/datasets/aymansharara/IdiomX

Kaggle:
https://www.kaggle.com/datasets/aymansharara/idiomx

The full data collection and enrichment pipeline is available at:

https://github.com/aymanshar/idiomx-dataset

Benchmark tasks, experiments, and evaluation scripts are provided at:

https://github.com/aymanshar/IdiomX

These resources include the dataset, preprocessing scripts, and task implementations, enabling straightforward integration and reproducibility.

X. Limitations

While IdiomX provides large-scale coverage and rich annotations, several limitations should be considered.

First, part of the dataset is generated using LLM-based methods, which may introduce minor inconsistencies or stylistic biases. Second, the Arabic data primarily reflects standard written Arabic and may not fully capture dialectal variation.

In addition, some generated examples may reflect model-preferred phrasing rather than naturally occurring usage frequencies. For the generation task, evaluation relies primarily on exact match metrics, which may not fully capture semantic equivalence between idiomatic expressions.

Future work includes incorporating human validation and extending the dataset to additional languages and dialects.

XI. Conclusion

In this work, we introduced IdiomX, a large-scale bilingual dataset for idiomatic expression understanding, providing broad coverage, rich semantic annotations, and aligned English–Arabic representations.

Through comprehensive benchmarking, we showed that idiom understanding requires deep contextual modeling. Transformer-based models significantly outperform lexical approaches for idiom detection, while hybrid generation–retrieval methods achieve the best performance for context-to-idiom prediction.

For the cross-lingual Arabic context-to-idiom task, retrieval-based models are effective at capturing general semantic alignment but remain limited in precise idiomatic mapping. Fine-tuning substantially improves performance, yet challenges such as semantic ambiguity, ranking limitations, and cross-lingual variation persist.

Overall, the results demonstrate that idiom understanding is inherently complex and cannot be fully addressed by a single

modeling paradigm. Future work should explore hybrid architectures that integrate retrieval, generation, and cross-lingual reasoning to better capture idiomatic meaning across languages.

Future work includes extending IdiomX to additional languages using the same pipeline.

XII. References

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